



RESEARCH ARTICLE

REVISED

Automatic Vehicle Fueling System using PLC Controlled Robotic Arm - A Simulation Design

[version 2; peer review: 2 approved, 1 approved with reservations]

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Abstract
The objective of this research is to simulate an automatic fuelling system using a PLC LogixPro simulation. The system includes the “FASS” concept, which is Fast, Accurate, Safe and Simple, to allow car users to have an efficient fuel filling system.

The design concept consists of three processes – identification of the vehicle, payment, and filling with the fuel. The first process identifies the presence of the car by the in-floor weight sensors. The weight sensor identifies the car, locks it in position, and activates the payment system. The second process activates the payment system. After payment is completed, the fuel cap will be opened to enable the system to start filling the fuel. If the payment doesn't go through the car will be released, manual operation will be initialized, and the entire system will be reset. A timer is included in the payment section to process the payment. In the third process, the filling arm is extended to the car, the fuel cap is opened, the fuel pump is inserted into the tank, and fuel is directed into the tank. Once the tank is full, filling is stopped, the pump is ejected, the fuel cap is closed, and the arm returned back to its position. Thus, an automatic vehicle fuelling system is created to overcome the problems of poor safety and longer waiting time during peak hours.

The fuel cap is activated and deactivated by pressure and the sensor filler is stopped by a level sensor. The pump insert is activated and deactivated by a photosensor.

Keywords
PLC, LogixPro, Automatic Vehicle Fueling System, Ladder Logic, Pressure sensor, fuel pump, fuel cap, FASS

Open Peer Review

Approval Status ✓ ? ✓

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研究文章

REVISED

基于PLC控制的自动车辆加油系统机械臂——仿真设计

[版本2；同行评审：2人批准，1人有保留批准]Chitra Venugopal¹, Bhuvaneswari Thangavel²

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摘要
本研究旨在通过PLC LogixPro仿真实现自动加油系统。该系统包含“FASS”概念，即快速、准确、安全、简单，使汽车用户能够拥有高效的加油系统。

设计概念包括三个过程——车辆识别、支付和加油。第一个过程通过地内重量传感器识别车辆存在。重量传感器识别车辆，将其锁定到位，并激活支付系统。第二个过程激活支付系统。支付完成后，油箱盖打开，使系统能够开始加油。如果支付未通过，车辆将被释放，手动操作启动，整个系统重置。支付部分包含一个计时器来处理支付。在第三个过程中，加油臂伸向车辆，打开油箱盖，将油泵插入油箱，燃油导入油箱。一旦油箱满，停止加油，弹出油泵，关闭油箱盖，加油臂返回原位。由此，创建了一个自动加油系统，以解决高峰时段安全性差和等待时间长的问题。

燃油盖通过压力激活和停用，传感器加油口由液位传感器停止。泵芯由光电传感器激活和停用。

关键词
PLC, LogixPro, 自动车辆加油系统, 梯形逻辑, 压力传感器, 燃油泵, 燃油盖, FASS

开放同行评审

审批状态 ✓ ? ✓

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REVISED Amendments from Version 1

Title is updated.
The description is modified and displayed in the first sentence of the Abstract.
Acronyms RFID and GSM are expanded in the first use.
Objective of this research is added at the end of section 1.
Section 2 Title is modified to System Design and 2.1 Design of the system is removed.
New figures Fig. 1 and 2 are added to explain the model and the algorithm.
Table 1 and Table 2 contents are moved after the Tables.
In Section 2, the figure legends are included and described in the text.
In Section 4, suggested future modifications are added at the end of the section.

Any further responses from the reviewers can be found at the end of the article

1. Introduction

The COVID-19 pandemic has put a spotlight on the role of automation in businesses. Since human contact spreads the disease, machines now have a protective role. This has accelerated the number of robots in industrial applications. The number of vehicles on roads is increasing every day. As fuel stations are operated manually delays are caused and there are long queues, especially in front of gas pumps. During fuelling, drivers must expose themselves to extreme hot or cold temperatures and may also come into contact with dangerous fumes. Thus, an automatic vehicle fuelling system is proposed to overcome problems like poor safety, longer waiting times during peak hours, and the spread of COVID-19.

Petroleum products are expensive. Hence the proper use and distribution of petroleum products is an important task in industries. Fuel stations are built to dispense gasoline and diesel into vehicles for commercial and non-commercial purposes.

In a study by Edward¹ to automate the fuel pumping system, an Radio Frequency Identification (RFID) based automated petrol pump system was proposed. In this, the RFID system is used to implement the design tasks sequentially. In Ref. [2](#) an inventory management procedure is proposed by which the supervisor can find the detailed information of the entity. They can delete the record if they want. Several modules (product cost, staff management) and report forms, like daily pump reports, shift delivery reports, cumulative daily reports, and salary reports are incorporated into the proposed system. It has a password facility for each module for safety purposes.

Fuel stations are operated manually in Bangladesh, leading to delays. Ali Newaz Bahar et al. proposed an automated fuel management system that can maintain the account of the fuel stations to reduce corruption in transactions.³ It prints a receipt automatically after every transaction and can monitor the transactions from a remote place via the internet.⁴ presents an RFID technology-based fuel dispensing system. The proposed system prevents unauthorized fueling in Baghdad city. Based on the type of vehicle, each registered vehicle will have a specific amount of fuel.

In Ref. [5](#), a Raspberry Pi based automated fuel pumping system is analyzed. The proposed system aims to confirm the resetting of the transaction every time the fuel pump nozzle is placed back on the dispenser station. In this design, a Raspberry Pi camera and Bluetooth communication are used to reset the dispenser unit when the vehicle passes the dispenser unit. The results show the successful implementation of the automation process.

M.Z.A Rashid et al. proposed an Automatic vehicle fuelling system that utilizes a positioning robot arm.⁶ They explored a new fuel dispensing system that allowed the robotic arm to move using its search head and extendable nozzle toward the fuel spot of the car sensors are used to identify the location and fuelling position of the vehicle.

In Ref. [7](#) Wavekar Asrar A et al. aims to design an electronic payment system by creating a prepaid card for petrol stations. In this design, RFID technology is used to automatically pay for petrol dispensed. In Ref. [8](#) a fuel filling automation process using RFID and Global System for Mobile Communication (GSM) technology is discussed. This design uses an RFID card and card reader. When the user swipes the RFID card in the card reader installed in the fuel station, it reads the card information, such as the quantity of fuel to be dispensed, and calculates the amount to deduct from the card. Upon receiving confirmation from the user and the amount remaining in the card, the amount is deducted, and the fuel is dispensed. The GSM technology is used to facilitate the online recharge facility.

Aishwarya Jadhav et al. design a prepaid card for petrol bunk systems and also petrol dispensing systems using RFID technology.⁹ Use of an unmanned power pump requires less time to operate, is effective, and can be installed anywhere.

修订版对版本1的修正

标题已更新。
描述已修改并显示在摘要的第一句。首字母缩写RFID和GSM在首次使用时已展开。本研究的目标已添加至第1节末尾。第2节标题修改为系统设计，并删除了2.1系统设计。
新增图1和图2以解释模型和算法。表1和表2的内容已移至表格之后。在第2节中，图例已包含并在文本中描述。在第4节中，建议的未来修改已添加至该节末尾。

审稿人的任何进一步回复可在文章末尾找到

1. 引言

COVID-19疫情凸显了自动化在商业中的作用。由于人际接触会传播疾病，机器现在承担了保护性角色。这加速了工业应用中机器人的数量。道路上的车辆数量每天都在增加。由于加油站人工操作，造成延误，尤其是在加油泵前排队很长。加油时，司机必须暴露在极端炎热或寒冷的温度下，还可能接触危险烟雾。因此，提出了自动车辆加油系统，以解决安全性差、高峰时段等待时间长以及COVID-19传播等问题。

石油产品价格昂贵。因此，正确使用和分配石油产品是工业中的一项重要任务。加油站是为商业和非商业目的向车辆分配汽油和柴油而建造的。

在Edward的一项研究中¹为了自动化燃油泵送系统，提出了一种基于射频识别（RFID）的自动化加油系统。在该系统中，RFID系统被用于顺序执行设计任务。在文献²中，提出了一种库存管理程序，管理员可以查找实体的详细信息，也可以根据需要删除记录。系统中融入了多个模块（如产品成本、员工管理）和报表形式，如每日加油报表、班次交付报表、累计日报表和工资报表。每个模块都有密码保护以确保安全。

孟加拉国的加油站通常由人工操作，导致延误。Ali Newaz Bahar等人提出了一种自动化燃油管理系统，该系统可以维护加油站的账目，以减少交易中的腐败行为。³ 每次交易后自动打印收据，并可通过互联网远程监控交易。⁴提出了一种基于RFID技术的燃油分配系统。该系统可防止巴格达市的非法加油。根据车辆类型，每辆注册车辆将获得特定量的燃油。

在参考文献⁵中，分析了一种基于树莓派的自动燃油泵系统。该系统旨在确认每次燃油泵喷嘴放回加油机时交易重置。该设计中，使用树莓派摄像头和蓝牙通信在车辆经过加油机时重置加油机。结果显示自动化过程成功实施。

M.Z.A Rashid等人提出了一种利用定位机器人手臂的自动车辆加油系统。⁶ 他们探索了一种新的燃油分配系统，允许机器人手臂使用其搜索头和可伸缩喷嘴朝向车辆的燃油点移动，通过传感器识别车辆的位置和加油位置。

在参考文献⁷中，Wavekar Asrar A等人旨在通过创建加油站预付卡来设计电子支付系统。该设计中，使用RFID技术自动支付分配的燃油。在参考文献⁸中，讨论了使用RFID和全球移动通信系统（GSM）技术的燃油加注自动化过程。该设计使用RFID卡和读卡器。当用户在加油站的读卡器上刷卡时，读取卡片信息，如要分配的燃油量，并计算从卡中扣除的金额。收到用户确认且卡内余额足够后，扣除金额并分配燃油。GSM技术用于支持在线充值功能。

Aishwarya Jadhav等人设计了一种用于加油站系统以及基于RFID技术的汽油分配系统的预付费卡。⁹ 使用无人值守的电动泵操作时间更短、效率高，且可安装在任何地方。

Since fuel stations are operated manually, they are time-consuming and require more manpower. In Ref. 10 a self-operated petrol pump is proposed. The use of unmanned petrol pump requires less time to operate. It is effective and can be installed anywhere the customer self-going to avail the services the payment is done by electronic clearing system.

M.B. Pranto et al. Proposed an RFID based secure fuel monitoring method with low labour cost. The goal of this system was to avoid the dishonesty of pump labourers & car drivers toward car owners.¹¹ This was achieved by developing a mobile application to trace a vehicle's fuel refill amount, refill cost, current and previous balance of the account & time of transaction. The users need to register themselves first to control their accounts.

In Ref. 12 an IoT-based automated petrol pump system for remote areas is proposed to advantage the petrol station owners. With this proposed smart petrol pump there is no need for a physical operator/person for the distribution of petrol. When a user needs to fill with fuel the vehicle first checks for pricing online in that specific portal designed for this petrol station. Then by selecting the nearby stations and pay for petrol amount. Then he proceeds go the station for filling.

In this research paper, development of PLC algorithm for automatic fuel filling system is discussed. The algorithm is divided into 3 sections – first section is to identify the presence and absence of the car and its position, second section is to process the payment system and the third section is to fuel filling. The simulation is performed using LogixPro simulation platform and the result of each section is presented. This research can be further developed by adding more features to the robotic arm such as windscreen cleaning and adding visible and audible alarms for each section.

2. System design

The functional diagram of the automatic fuelling system is shown in Figure 1. The functional diagram is divided into three major sections – identification of the vehicle, payment confirmation, and fuel filling which are activated by the sensor inputs and controlled using PLC LogixPro programming.

The system is designed with a robotic arm. The armrests are in position and get activated when the car is parked in the designated space in front of the pumping station. The in-floor weight sensor is activated by the presence of the car which sends a signal to the PLC system indicating that the car has arrived at this pumping station. The PLC system activates the robotic arm and the payment system. Once the payment is successful, the robotic arm is extended to the tank position to fill the fuel. A photoelectric sensor (1) is used to track the position of the arm. When the arm is released from its position, the photoelectric sensor output sends a signal to the PLC to activate the pump insert motor. The opening of the fuel cap is identified by the photoelectric sensor (2). An ultrasonic sensor is used to identify the distance reached by the robotic arm to make sure that the arm has reached the tank position. The ultrasonic sensor and two photoelectric sensors are used to activate the pump insert motor. Once the pump is inserted into the tank, fuel filling happens until the tank level is reached. When the fuel filling is stopped, the pump eject motor is activated, to release the pump from the tank, and the fuel tank cap is closed, the arm is returned back to its original position and is ready for the next fuel filling.

The following algorithm explains the program which controls the entire automation operation.

- The built-in in-floor weight sensor senses the car and activates the payment sensor.
- Once the payment is completed, the arm is extended for fuel filling, if the payment is not completed, the system will be reset, and manual operation will be initiated.



Figure 1. Functional diagram of the automatic fuel filling system.

由于加油站是人工操作，因此耗时且需要更多人力。在文献10中，提出了一种自助式汽油泵。使用无人汽油泵所需操作时间更少。它非常有效，可以安装在任何客户自行前往享受服务的地方，付款通过电子清算系统完成。

M.B. Pranto 等人提出了一种基于 RFID 的安全燃料监测方法，劳动力成本低。该系统的目标是避免加油站工人和汽车司机对车主的不诚实行为。¹¹ 这是通过开发一个移动应用程序来实现的，该应用程序可以追踪车辆的加油量、加油费用、账户当前和之前的余额以及交易时间。用户需要先注册才能控制自己的账户。

在参考文献12中，提出了一种用于偏远地区的基于物联网的自动加油站系统，以造福加油站老板。通过这种提议的智能加油站，无需物理操作员/人员来分配汽油。当用户需要加油时，车辆首先在该加油站专门设计的门户网站上在线查询价格。然后通过选择附近的加油站并支付油费。之后他前往加油站进行加油。

本论文讨论了用于自动加油系统的 PLC 算法的开发。该算法分为三个部分：第一部分是识别车辆的存在与否及其位置，第二部分是处理支付系统，第三部分是加油。使用 LogixPro 仿真平台进行仿真，并展示了每个部分的结果。这项研究可以通过为机械臂增加更多功能（如挡风玻璃清洁）以及为每个部分添加可见和可听的警报来进一步开发。

2. 系统设计

自动加油系统的功能图如图1所示。功能图分为三个主要部分——车辆识别、支付确认和燃油加注，这些部分由传感器输入激活，并使用PLC LogixPro编程进行控制。

该系统设计有机械臂。当汽车停放在加油站的指定位置时，扶手处于就绪状态并被激活。地埋式重量传感器检测到汽车后，向PLC系统发送信号，表明汽车已到达该加油站。PLC系统激活机械臂和支付系统。一旦支付成功，机械臂伸展到油箱位置进行加油。使用光电传感器（1）跟踪机械臂的位置。当机械臂脱离其位置时，光电传感器输出信号给PLC，激活油泵插入电机。光电传感器（2）用于识别油箱盖是否打开。超声波传感器用于识别机械臂到达的距离，确保机械臂已到达油箱位置。超声波传感器和两个光电传感器用于激活油泵插入电机。一旦油泵插入油箱，便开始加油，直到达到油箱液位。加油停止后，激活油泵退出电机，将油泵从油箱中退出，关闭油箱盖，机械臂返回原始位置，准备下一次加油。

以下算法说明了控制整个自动化运行的程序。

- 内置的地板重量传感器检测到车辆并激活支付传感器。
- 一旦支付完成，机械臂伸出进行加油；如果支付未完成，系统将重置，并启动手动操作。



图1. 自动加油系统功能框图。

- Fuel arm position sensor is used to activate the fuel arm motor
- Fuel arm position sensor and fuel arm and fuel arm motor activate the fuel cap opener.
- Fuel cap open sensor is used to activate the fuel pump insert motor
- A fuel pump insert sensor is used to start the fuel filling system
- The level sensor stops the filling once the tank reaches the full level
- The stop full sensor will eject the pump, the fuel cap will close, the arm will return to its position and the car will be released to move.

The block diagram and the flowchart of the algorithm is shown in Figure 2.

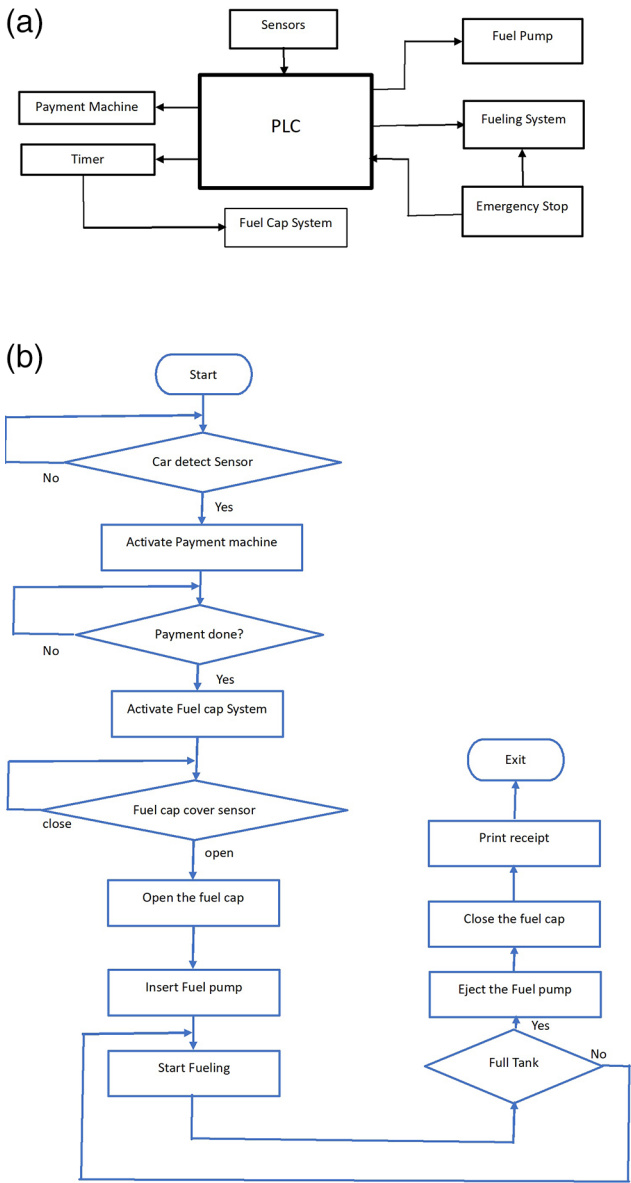


Figure 2. Block diagram and algorithm of the design.

- 燃油臂位置传感器用于激活燃油臂电机• 燃油臂位置传感器、燃油臂和燃油臂电机激活油箱盖开启器• 油箱盖开启传感器用于激活燃油泵插入电机• 燃油泵插入传感器用于启动加油系统• 液位传感器在油箱达到满位时停止加油• 满位传感器会弹出油泵，关闭油箱盖，机械臂返回原位，车辆被释放移动。

算法的框图和流程图如图2所示。

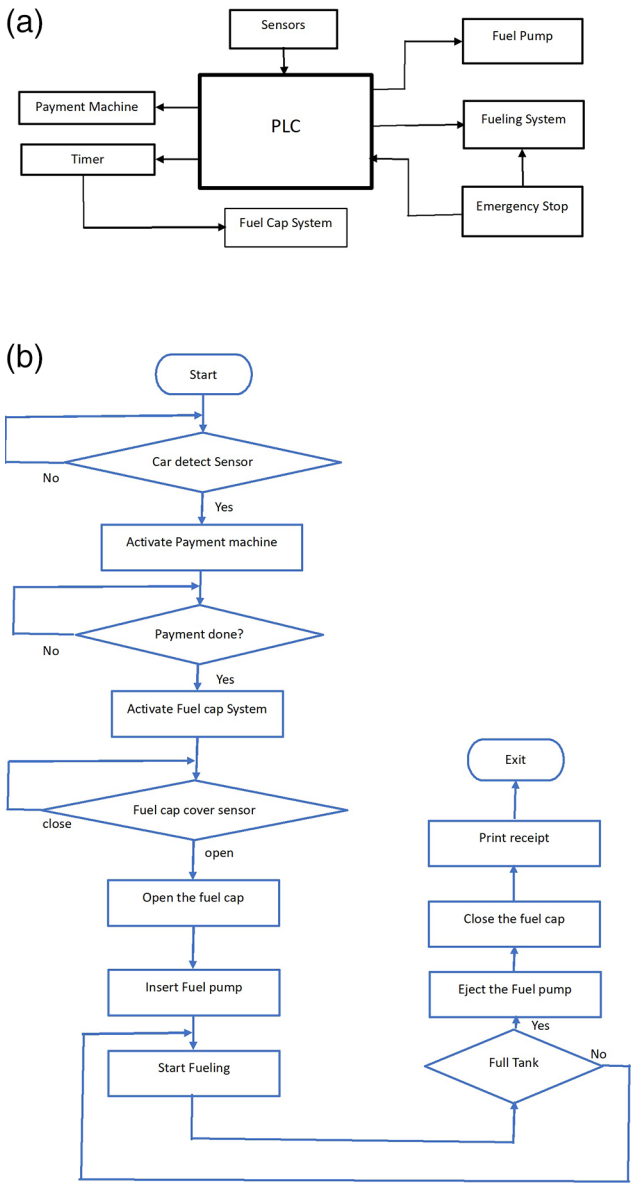


图2.设计的框图和算法。

Table 1. Input components along with their addresses.

Operation	Type of device	Port address
Stop	Normally Closed Push Button	I:1/0
Start	Normally Open Push Button	I:3/0
Car Sensor	Weight Sensor	I:1/1
Payment Done	Automatic Signal from the Payment system	I:1/2
Manual Input	Normally open Push Button for Manual Payment system	I:1/4
Fuel Arm Position detector	Ultrasonic Sensor to calculate the distance	I:1/3
Fuel Arm position sensor	Photoelectric sensor 1	I:1/6
Fuel cap sensor	Photoelectric sensor 2	I:1/7
Pump insert sensor	Limit switch	I:1/8
Full level sensor	Fluid level switch	I:1/9

Table 2. Output components along with their addresses.

Operation	Type of device	Port address
Lock the Car	Push Button Switch	O:2/0
Payment system	Motor	O:2/1
Fuel arm motor	Motor	O:2/2
Release the Car	Push Button Switch	O:2/3
Fuel cap opener	Motor	O:2/4
Fuel cap close	Motor	O:2/8
Pump insert motor	Motor	O:2/5
Pump eject motor	Motor	O:2/7
Fill fuel	Limit switch	O:2/6

The input components along with their addresses are shown in Table 1. The output components, along with their addresses, are shown in Table 2.

3. Results and analysis

The PLC simulation program is simulated using LogixPro 500 (<https://www.plclogix.com>) I/O simulator switches.

Step 1: Activate the system – Sensing the car.

The car sensor is activated by the limit switch. When the car sensor is activated, it activates the payment system. The car sensor algorithm is shown in Figure 3a. The switch positions and the indicators are shown in Figure 3b.

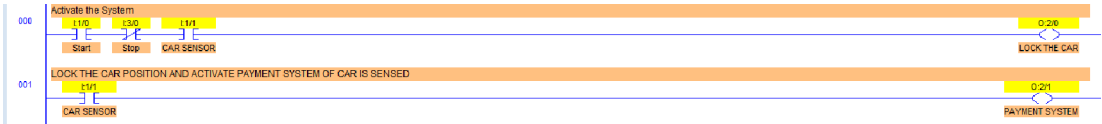


Figure 3a. Sensing the car.

表1.输入元件及其地址。

操作	设备类型	端口地址
Stop	常闭按钮	I:1/0
启动	常开按钮	I:3/0
车辆传感器	重量传感器	I:1/1
支付完成	来自支付系统的自动信号	I:1/2
手动输入	手动支付系统的常开按钮	I:1/4
油臂位置检测器	超声波传感器，用于计算距离	I:1/3
油臂位置传感器	光电传感器1	I:1/6
油箱盖传感器	光电传感器2	I:1/7
油枪插入传感器	限位开关	I:1/8
满液位传感器	液位开关	I:1/9

表2. 输出组件及其地址。

操作	设备类型	端口地址
锁定汽车	按钮开关	O:2/0
支付系统	电机	O:2/1
加油臂电机	电机	O:2/2
释放车辆	按钮开关	O:2/3
油箱盖开启器	电机	O:2/4
油箱盖关闭	电机	O:2/8
泵插入电机	电机	O:2/5
泵弹出电机	电机	O:2/7
加注燃油	限位开关	O:2/6

输入元件及其地址如表1所示。输出元件及其地址如表2所示。

3. 结果与分析

PLC 仿真程序使用 LogixPro 500 (<https://www.plclogix.com>) I/O 模拟器开关进行仿真 s.

步骤 1：激活系统 – 感应车辆。

车辆传感器 由限位开关激活。当车辆传感器被激活时，它会启动支付系统。汽车传感器算法 算法如图3a所示。开关位置和指示灯如图3b所示。 r

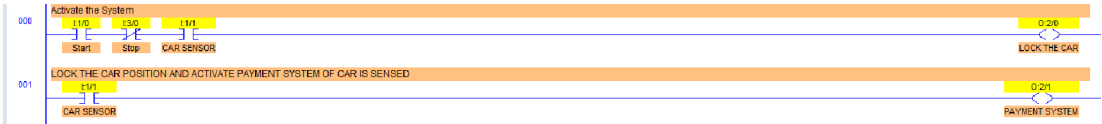


图3a. 感应车辆。

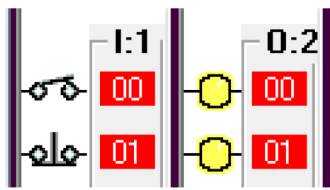


Figure 3b. Switch position and indicators.

Step 2: The payment system activates the timer. The algorithm is shown in Figure 3c.



Figure 3c. Timer for activating payment system.

Step 3: When the payment is done at or before the timer bit is done, the fuel arm motor is activated.

To simulate the system, the ‘payment done’ signal is generated by using a push-button switch in the I/O simulator. To ensure the safety of the system, before activating the fuel arm motor, the fuel arm position detector, pump eject motor, fuel cap close sensor and car release sensor are disabled. Also, manual input is added to bypass the automatic system in case of any error in the automatic system.

The algorithm related to this action is shown in Figure 3d. The sensor status and the indicators are shown in Figure 3e.

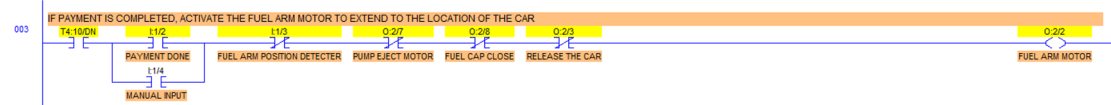


Figure 3d. Payment complete.

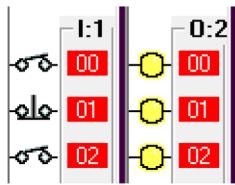


Figure 3e. Payment complete sensor and indicators.

Step 4: If the payment doesn’t go through then release the car and reset the system.

To simulate this condition, the ‘payment not done’ signal is activated using a push-button switch in the I/O simulator. The algorithm and sensor status is given in Figures 3f and 3g respectively.



Figure 3f. Payment failure action.

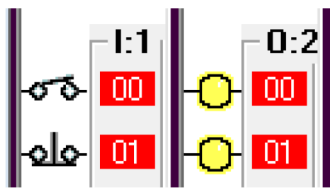


图3b. 开关位置和指示灯。

步骤2：支付系统启动计时器。算法如图3c所示。



图3c. 用于激活支付系统的定时器。

步骤3：在计时器位完成之前或之时付款，燃油臂电机被激活。

为了模拟系统，使用I/O模拟器中的按钮开关生成“付款完成”信号。为确保系统安全，在激活燃油臂电机之前，燃油臂位置检测器、泵喷射电机、油箱盖关闭传感器和汽车释放传感器均被禁用。此外，添加了手动输入，以便在自动系统出现任何错误时绕过自动系统。

该算法 与此动作相关的算法如图3d所示。传感器状态和指示灯如图3所示。 e.

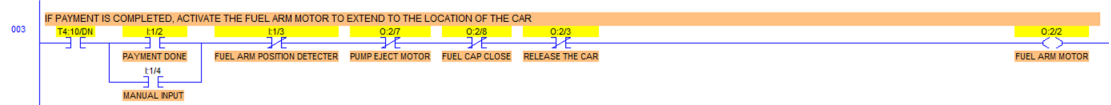


图3d. 支付完成。

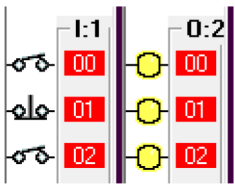


图3e. 支付完成传感器和指示灯。

步骤4：如果付款未通过，则释放汽车并重置系统。

为了模拟这种情况，使用I/O模拟器中的按钮开关激活“付款未完成”信号。算法和传感器状态分别如图3f和3g所示。



图 3f. 支付失败操作。

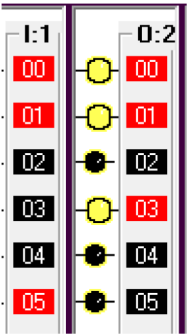


Figure 3g. Sensor and indicator signals for payment failure.

Step 5: Open fuel cap.

To simulate this condition, the fuel arm position sensor is simulated using a switch in the I/O simulator. For the safety of the system, the fuel cap close condition is used as normally closed in this statement. If the fuel cap is already opened, then the fuel cap opener will not operate. The fuel cap opening is shown in Figure 3h. The sensor and indicator signals are shown in Figure 3i.



Figure 3h. Fuel cap opener.

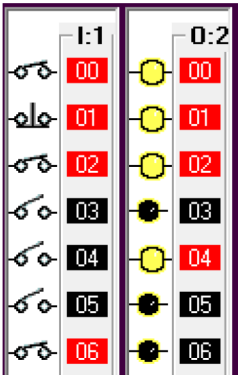


Figure 3i. Sensor and indicator status for fuel cap opener.

Step 6: Insert pump.

The fuel cap sensor senses the opening of the fuel cap of the car tank. The fuel cap sensor and fuel cap opener activate the fuel arm insert motor. To simulate this step, the fuel cap sensor input is simulated by using a switch in the I/O simulator. The algorithm activating the fuel pump insert motor is shown in Figure 3j. The sensor status and indicators are shown in Figure 3k.



Figure 3j. Fuel pump insert motor.

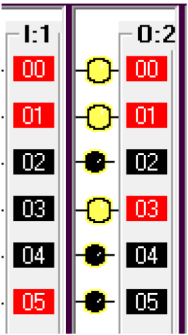


图3g. 支付失败的传感器和指示器信号。

步骤 5：打开燃油盖。

为了模拟这种情况，使用I/O模拟器中的开关来模拟燃油臂位置传感器。为系统安全起见，在此语句中，油箱盖关闭状态使用常闭触点。如果油箱盖已打开，则油箱盖开启器不会操作。油箱盖开启情况如图3h所示。传感器和指示器信号如图3i所示。



图3h。 燃油盖开启器。

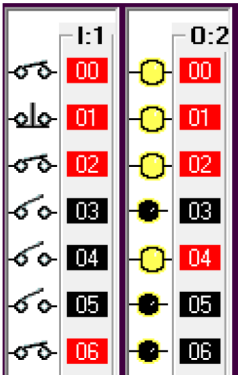


图3i. 油箱盖开启器的传感器和指示器状态。

步骤6：插入油泵。

燃油盖传感器感应汽车油箱盖的开启。燃油盖传感器和燃油盖开启器激活燃油臂插入电机。为模拟此步骤，通过I/O模拟器中的开关模拟燃油盖传感器输入。激活燃油泵插入电机的算法如图3j所示。传感器状态和指示器如图3k所示。



图 3j. 燃油泵插入电机。

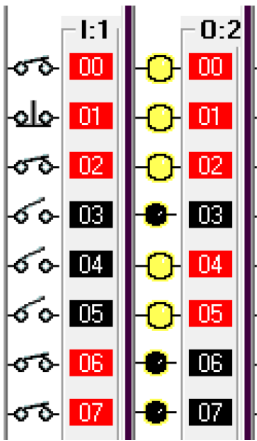


Figure 3k. Sensor and indicator status for fuel pump insert motor.

Step 7: Fill fuel.

Once the pump is inserted into the tank, it will activate the limit switch which is the pump insert sensor. The full level sensor is a liquid level sensor used to identify the fuel level of the tank. It is used as a normally closed switch as a safety condition to avoid overflow. The pump insert sensor, full level sensor and pump insert motor condition activate the fuel fill start operation. To simulate this condition, the pump insert sensor, is input is used using a I/O simulator switch. The fuel filling algorithm is shown in Figure 3l. The sensor and status indicators are shown in Figure 3m.



Figure 3l. Fuel fill activation.

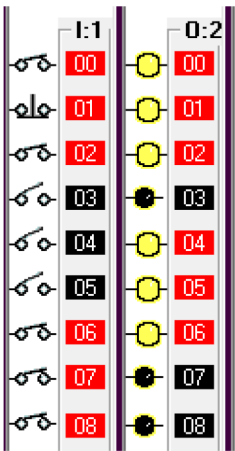


Figure 3m. Sensor and indicator status for fuel filling.

Step 8: Pump eject.

Once the full level is reached, the pump eject motor is activated. The full level sensor is simulated using an I/O simulator switch. The pump eject motor algorithm is shown in Figure 3n. The corresponding sensor and indicator status is shown in Figure 3o.

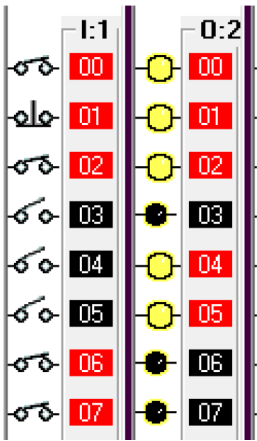


图3k. 油泵插入电机的传感器和指示器状态。

步骤 7：加注燃油。

一旦油泵插入油箱，它会激活限位开关，即油泵插入传感器。满液位传感器是一种液位传感器，用于识别油箱的燃油液位。它作为常闭开关使用，作为避免溢出的安全条件。油泵插入传感器、满液位传感器和油泵插入电机状态激活燃油加注启动操作。为了模拟这一条件，油泵插入传感器的输入通过I/O模拟器开关使用。燃油加注算法如图3l所示。传感器和状态指示器如图3m所示。



图3l. 燃油加注激活。

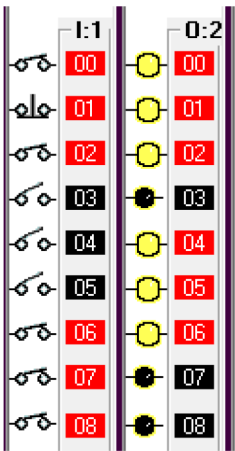


图3m. 燃油加注过程中的传感器和指示器状态。

步骤8：油泵弹出。

一旦达到满液位，泵排出电机即被激活。满液位传感器通过I/O模拟器开关模拟。泵排出电机算法如图3n所示。相应的传感器和指示灯状态如图3o所示。

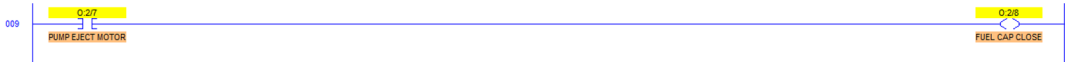


Figure 3n. Pump eject and fuel cap close.

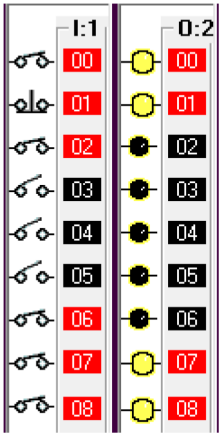


Figure 3o. Sensor and indicator status for fuel cap close.

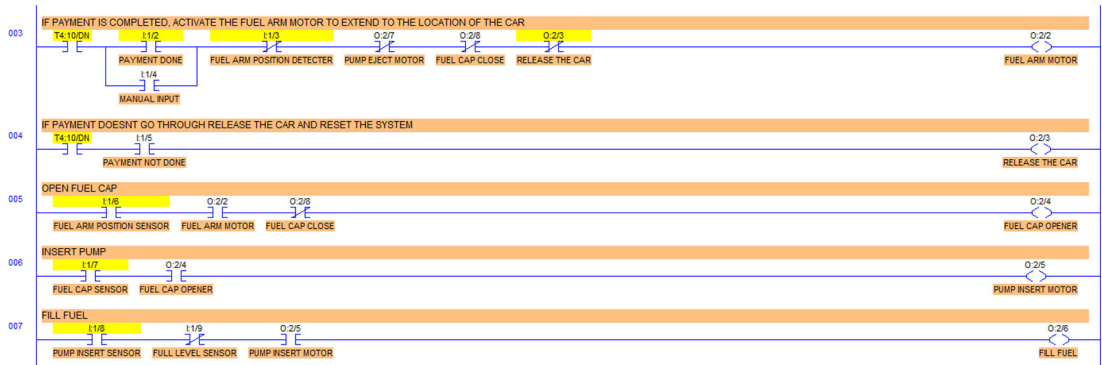


Figure 3p. Failsafe method.

This condition deactivates the pump insert motor, fuel cap opener, fuel arm motor and returns the robotic arm to its original position and releases the car. This can be seen from the deactivated output O:2/2, O:2/3, O:2/4, O:2/4, O:2/5, O:2/6. This is shown in Figure 3p.

Thus, the algorithm for the automatic fuelling system works in the failsafe mode and achieves the objective successfully.

4. Conclusion

The application of PLC in industry is inevitable and is unexplored in the context of fuel filling stations so far. To avoid crimes happening in fuel filling stations due to manual operation and delays caused by manual operation, this automatic fuel filling is suggested. Also, the recent pandemic situation necessitates reducing physical distancing and exchanging of payment systems between customers due to which many fuel stations were closed during pandemic times. Also, the long waiting times during peak hours due to manual operation indicate a dire need for automatic fuel filling systems in this field of engineering. The design of the system considered for this study is explained in this paper. The programming was developed using LogixPro simulation and tested using the I/O simulator. The step-by-step test results were explained. The study shows that the algorithm was implemented successfully and is ready for hardware implementation.

The system can be upgraded by adding visual and audible signal at each stages of fuel filling processing. Also additional features such as cleaning the windscreen can be added to the robotic arm.

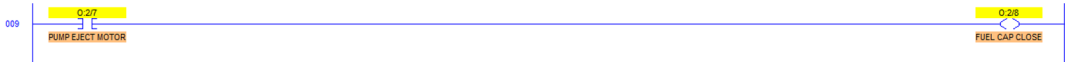


图3n. 泵弹出并关闭油箱盖。

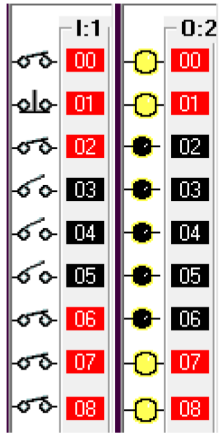


图3o. 燃油盖关闭时的传感器和指示灯状态。

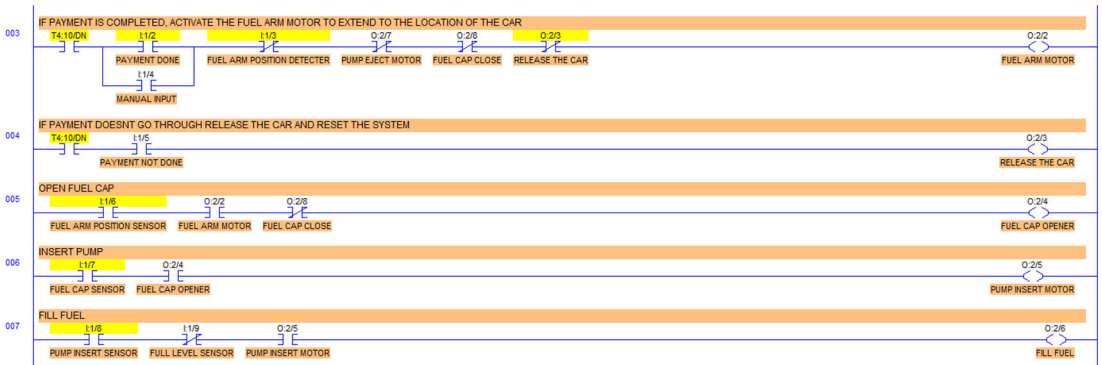


图3p. 故障安全方法。

此状态停用泵插入电机、油箱盖开启器、燃油臂电机，并将机械臂返回原始位置并释放汽车。这可以从停用的输出O:2/2、O:2/3、O:2/4、O:2/4、O:2/5、O:2/6看出。如图3p所示。

因此，该算法用于自动加油系统，在故障安全模式下工作，并成功实现了目标。

4. 结论

PLC在工业中的应用是不可避免的，并且在加油站的背景下至今尚未被探索。为了避免因人工操作导致的加油站犯罪和延误，本文提出了自动加油方案。此外，最近的疫情要求减少人际接触和支付系统的交换，导致许多加油站疫情期间关闭。同时，高峰时段因人工操作而导致的长时间等待表明，该工程领域迫切需要自动加油系统。本文解释了我们研究的系统设计。使用LogixPro仿真开发了程序，并通过I/O模拟器进行了测试。逐步解释了测试结果。研究表明，该算法已成功实现，并准备好进行硬件实施。

该系统可通过在加油过程的各个阶段添加视觉和听觉信号进行升级。此外，还可以为机械臂增加清洁挡风玻璃等额外功能。

Data availability statement

All data underlying the results are available as part of the article and no additional source data are required.

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数据可用性声明

所有结果背后的数据均作为文章的一部分提供，无需额外源数据。

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Open Peer Review

Current Peer Review Status: ✓ ? ✓

Version 2

Reviewer Report 07 May 2024

<https://doi.org/10.5256/f1000research.164528.r270334>

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The changes made as a result of the previous revision have significantly improved the quality and presentation of the content of the manuscript.

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Microgrids, Hydrogen, PEM Electrolyzers, Photovoltaic, Renewable Energies, Automation, Internet of Things, Industrial IoT.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Version 1

Reviewer Report 18 April 2023

<https://doi.org/10.5256/f1000research.77340.r166988>

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✓ **Md. Tabil Ahammed** 
¹ Department of ECE, Khulna University of Engineering and Technology, Khulna, Bangladesh
² Department of ECE, Khulna University of Engineering and Technology, Khulna, Bangladesh

Great work on designing and developing a robotic arm for an automatic fueling system using a

开放同行评审

当前同行评审状态: ✓ ? ✓

版本 2

审稿报告 2024年5月7日

<https://doi.org/10.5256/f1000research.164528.r270334>

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✓ **Francisco Javier Folgado** 
西班牙埃斯特雷马杜拉大学，埃斯特雷马杜拉，西班牙

根据上次修订所做的更改显著提高了稿件内容的质量和呈现效果。

利益冲突：未披露任何利益冲突。

审稿人专长：微电网、氢能、PEM电解槽、光伏、可再生能源、自动化、物联网、工业物联网。

我确认已阅读此投稿，并相信我有足够的专业水平确认其具有可接受的科学标准。

版本1

审稿人报告 2023年4月18日

<https://doi.org/10.5256/f1000research.77340.r166988>

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✓ **Md. Tabil Ahammed** 
1 孟加拉国库尔纳工程技术大学电子与通信工程系，库尔纳，孟加拉国 2 孟加拉国库尔纳工程技术大学电子与通信工程系，库尔纳，孟加拉国

在使用PLC LogixPro模拟设计开发自动加油系统的机械臂方面，您做得非常出色。

PLC LogixPro simulation. Your use of the FASS concept to ensure that the system is Fast, Accurate, Safe and Simple is commendable. The three-step process you have identified for identification, payment, and fuel filling is well thought out and ensures a smooth user experience. The inclusion of safety features such as the weight sensor, pressure-activated fuel cap, level sensor for filler, and photosensor for pump insert are impressive. This automatic vehicle fueling system is an innovative solution to overcome the problems of poor safety and longer waiting times during peak hours. Overall, this is a well-written report.

Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

Yes

Are sufficient details of methods and analysis provided to allow replication by others?

Partly

If applicable, is the statistical analysis and its interpretation appropriate?

Yes

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Photonics, Optoelectronics, WSN, Renewable Energy, ML

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Author Response 01 Apr 2024

Chitra Venugopal

Thank you. The report is modified with 2 new figures to help reader understand the concept better.

Competing Interests: No competing interests were disclosed.

您运用FASS理念确保系统快速、准确、安全、简便，值得称赞。您确定的识别、支付和加油三步流程经过深思熟虑，确保了流畅的用户体验。安全功能的加入，如重量传感器、压力感应油箱盖、加油口液位传感器和泵插入光电传感器，令人印象深刻。该自动车辆加油系统是一项创新解决方案，可克服高峰时段安全性差和等待时间长的问题。总体而言，这是一份撰写良好的报告。

工作是否清晰准确地呈现，并引用了当前文献？ 是

研究设计是否合适，技术上是否可靠？ 是

是否提供了足够的方法和分析细节，以便他人重复？ 部分

如果适用，统计分析及其解释是否恰当？ 是

结果的所有源数据是否可用以确保完全可重复性？ 是

结论是否得到结果的充分支持？ 是

竞争利益：未披露任何竞争利益。

审稿人专长：光子学、光电子学、无线传感器网络、可再生能源、机器学习

我确认已阅读此提交，并认为我有足够的专业知识确认其达到可接受的科学标准。

作者回复 2024年4月1日

Chitra Venugopal

谢谢。报告已修改，新增了2张新图以帮助读者更好地理解概念。

利益冲突声明：未披露任何利益冲突。

审稿人报告 2023年4月11日

<https://doi.org/10.5256/f1000research.77340.r166991>

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S. V. Viraktamath 

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The paper requires quite a few modifications and the manuscript should be in order to the required format to be accepted. Some recommendations are listed below:

- The contents in the Abstract need to be as per the given format specified. Abbreviations needs to be specified along with their full forms.
- The acronyms must be defined initially in the Section 1. The Literature survey carried out in the same section has various articles along with various approaches carried out doesn't give a comparative approach thus the survey does not give a summary.
- In the section 2, since there is just one subsection, no need to make a separate section 2.1. Objective of the work carried out in the manuscript is missing.
- The figures in the manuscript does not have any figure numbers along with their description. Also, the figure explanation along with their numbers should be carried out in the content that is followed.

Is the work clearly and accurately presented and does it cite the current literature?
Yes

Is the study design appropriate and is the work technically sound?
Partly

Are sufficient details of methods and analysis provided to allow replication by others?
Partly

If applicable, is the statistical analysis and its interpretation appropriate?
Partly

Are all the source data underlying the results available to ensure full reproducibility?
Partly

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

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S. V. Viraktamath 

1 印度达尔瓦德SDM工程技术学院电子与通信工程系

2 印度达尔瓦德SDM工程技术学院电子与通信工程系

本文需要较多修改，稿件应按要求的格式整理才能被接受。以下是一些建议：

摘要中的内容需按指定格式编写。缩写 需要注明其全称。

缩写必须在第1节中首次出现时定义。在第1节中进行的文献综述包含了多篇文章和多种方法，但未提供比较性分析，因此综述没有给出总结。

第2节中，由于只有一个小节，因此无需单独设立2.1节。 论文中缺少所开展工作目标的说明。

稿件中的图没有与描述对应的图号。此外，应在后续内容中给出带图号的图注。

工作是否清晰准确地呈现，并引用了当前文献？ 是

研究设计是否合适，技术上是否可靠？ 部分

是否提供了足够的方法和分析细节，以便他人重复？ 部分

如果适用，统计分析及其解释是否恰当？ 部分

结果背后的所有源数据是否可用以确保完全可重复性？ 部分

得出的结论是否得到结果充分支持？ 是

利益冲突：未披露任何利益冲突。

Reviewer Expertise: Error control coding and Image processing

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 01 Apr 2024

Chitra Venugopal

- 1. The description is modified and displayed in the first sentence of the Abstract
- 2. Acronyms RFID and GSM are expanded in the first use.
- 3. Objective of this research is added at the end of section 1.
- 4. Section 2 Title is modified to System Design and 2.1 Design of the system is removed
- 5. New figures fig.1 and 2 are added to explain the model and the algorithm
- 6. Table 1 and Table 2 contents are moved after the Tables
- 7. In Section 2, the figure legends are included and described in the text
- 8. In Section 4, suggested future modifications are added at the end of the section.

Competing Interests: No competing interests were disclosed.

Reviewer Report 13 February 2023

<https://doi.org/10.5256/f1000research.77340.r162216>

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✖ Francisco Javier Folgado 

- ¹ University of Extremadura, Extremadura, Spain
- ² University of Extremadura, Extremadura, Spain

The manuscript covers an interesting R&D topic and fits the scope of the Journal. Nonetheless, the paper requires substantial modifications to its content and format to improve its quality and presentation in order to be accepted. Some considerations and recommendations are listed below.

First of all, both the title of the paper and the content of the abstract do not accurately reflect the scope of the manuscript. In both, it is implied that all issues related to the development and design of the robotic arm will be addressed. However, the content of the manuscript only focuses on the development of the PLC control algorithm that will command the robot and its testing by means of a simulation. Therefore, the title of the document should be modified to clearly reflect

评审专家专长：错误控制编码与图像处理

我确认已阅读此提交，并相信我有足够的专业知识确认其具有可接受的科学标准，但我有如上所述的重大保留意见。

作者回复 2024年4月1日

Chitra Venugopal

- 描述已修改并显示在摘要1的第一句中。1. 首字母缩写RFID和GSM在首次使用时已展开。
- 2. 本研究的目的添加在第1节末尾。3. 第2节标题修改为系统设计，并删除了2.1系统设计。4. 添加了新图1和图2以解释模型和算法。5. 表1和表2的内容移至表格之后。6. 在第2节中，包含了图例并在文本中进行了描述。7. 在第4节中，建议的未来修改添加在该节末尾。8.

利益冲突声明：未披露任何利益冲突。

审稿报告 2023年2月13日

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✖ Francisco Javier Folgado 

- ¹ 西班牙埃斯特雷马杜拉大学，埃斯特雷马杜拉，西班牙
- ² 西班牙埃斯特雷马杜拉大学，埃斯特雷马杜拉，西班牙

该手稿涵盖了一个有趣的研发主题，符合期刊的范围。然而，论文需要对其内容和格式进行大量修改，以提高其质量和表达，才能被接受。以下是一些考虑和建议。

首先，论文的标题和摘要内容均未准确反映稿件范围。两者都暗示将涉及机械臂开发与设计的所有问题，但稿件内容仅关注通过仿真测试指挥机器人的PLC控制算法的开发。因此，应修改文档标题以清晰反映

the content of the paper.

In section 1. Introduction, the acronyms RFID and GSM must be defined before their use. Additionally, references used in the paper are cited in two different formats, a single format should be used for all references.

Continuing with the Introduction section, this section reviews various articles related to automatic refueling systems. They highlight key aspects of each one or the technologies used. Following this literature review, the reviewer recommends including a paragraph expressly describing the novel aspects of the research carried out. In addition, if appropriate, authors are encouraged to include a comparative table grouping the common aspects of the reviewed works, as well as this paper. In this way, all the aspects of the literature review as well as the novelty of the article are summarized and presented in a visual and simplified way.

A paragraph should be included detailing the objective and scope of the paper, clearly specifying that only the development of the control algorithm of the automatic refueling system by means of a PLC will be addressed, the results of which will be simulated.

A common practice in scientific papers consists on providing a brief overview of the structure of the paper at the end of the Introduction. This reviewer suggests including such information for a better presentation.

In section 2. Methods, the operation of the presented algorithm is described. The heading of subsection 2.1 "Design of the system" should be deleted, since there is no other subsection within section 2.

The content of this section repeatedly describes how the algorithm works. Firstly, by means of the second paragraph of the section. Secondly, by means of a step-by-step description of the algorithm. In order to eliminate this repetition and to enhance the explanation of the algorithm, the reviewer recommends eliminating the step-by-step description and replacing it with a flowchart-like figure that shows each of these steps in a connected and orderly manner.

The sentence " Operation and address" should be deleted.

It is recommended to relocate the phrase " The input components along with their addresses are shown in Table 1. The output components, along with their addresses, are shown in Table 2." before displaying the tables.

Section 3. Results and analysis presents the simulation process of the developed algorithm. On the other hand, all the figures shown in this section lack a title or legend. It is required to include an individual title for each of the figures presented in the paper.

Regarding section 4. Conclusions, it is highly recommended to include some sentences related to future work derived from the results obtained in this research.

Is the work clearly and accurately presented and does it cite the current literature?

No

论文内容。

在引言部分中，应在首次使用前定义缩略词RFID和GSM。此外，论文中引用的参考文献采用了两种不同的格式，应统一为一种格式。

继续引言部分，本部分回顾了与自动加油系统相关的多篇文章。它们突出了每篇文章的关键方面或所使用的技术。在文献综述之后，评审人建议增加一段明确描述所开展研究的创新点的内容。此外，如果合适，鼓励作者加入一个对比表格，汇总所综述文献以及本文的共同方面。这样，文献综述的所有方面以及文章的创新点都能以视觉化和简化的方式总结呈现。

应增加一段内容，详细说明本文的目标和范围，明确指定仅涉及通过PLC实现的自动加油系统控制算法的开发，其结果将通过仿真得到。

科学论文的常见做法是在引言末尾简要介绍论文的结构。本审稿人建议包含此类信息以更好地呈现。

在第2节“方法”中，描述了所提出算法的操作。应删除子节2.1“系统设计”的标题，因为第2节内没有其他子节。

本节内容反复描述了算法的工作原理：首先通过该节的第二段，然后通过算法的逐步描述。为消除此重复并增强对算法的解释，审稿人建议删除逐步描述，并用一个流程图式图形替换，以连贯有序的方式展示每个步骤。

句子“操作与地址”应删除。

建议将短语“输入组件及其地址如表1所示。输出组件及其地址如表2所示。”移至表格显示之前。

第3节“结果与分析”展示了所开发算法的仿真过程。此外，本节中显示的所有图表均缺少标题或图例。需要为论文中呈现的每个图表单独添加标题。

关于第4节“结论”，强烈建议添加一些与本研究结果相关的未来工作方向的句子。

工作是否清晰准确地呈现，并引用了当前文献？ 否

Is the study design appropriate and is the work technically sound?

Partly

Are sufficient details of methods and analysis provided to allow replication by others?

Partly

If applicable, is the statistical analysis and its interpretation appropriate?

Not applicable

Are all the source data underlying the results available to ensure full reproducibility?

No source data required

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Microgrids, Hydrogen, PEM Electrolyzers, Photovoltaic, Renewable Energies, Automation, Internet of Things, Industrial IoT.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

Author Response 01 Apr 2024

Chitra Venugopal

The suggested reviews are implemented in the revised version

Competing Interests: No competing interests were disclosed.

Author Response 01 Apr 2024

Chitra Venugopal

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8. In Section 4, suggested future modifications are added at the end of the section.

研究设计是否恰当，技术上是否合理？ 部分

是否提供了足够详细的方法和分析，以便他人复制？ 部分

如果适用，统计分析及其解释是否恰当？ 不适用

结果所依据的所有源数据是否均可获取以确保完全可重复性？ 无需源数据

得出的结论是否得到结果的充分支持？ 部分

利益冲突： 未披露任何利益冲突。

审稿人专长：微电网、氢能、PEM电解槽、光伏、可再生能源、自动化、物联网、工业物联网。

我确认已阅读此提交，并认为我有足够的专业知识来声明，基于上述原因，我不认为它达到可接受的科学标准。

作者回复 2024年4月1日

Chitra Venugopal

建议的审阅意见已在修订版本中实施。

利益冲突： 未披露任何利益冲突。

作者回复 2024年4月1日

Chitra Venugopal

描述已修改并显示在摘要1的第一句中。缩略词RFID和GSM在首次使用时展开。2. 研究目标添加在第1节末尾。3. 第2节标题修改为系统设计，并删除了2.1系统设计。4. 添加新图Fig.1和2以解释模型和算法。5. 表1和表2的内容移至表格之后。6. 在第2节中，图例已包含并在文本中描述。7. 在第4节中，建议的未来修改添加在该节末尾。8.

Competing Interests: No competing interests were disclosed.

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